Textbook — Notes

Boas: *Mathematical Methods in the Physical Sciences*

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Abstract

Problems involving vibrations or oscillations occur freuquently in physics and engineering. The Fourier series is a tool used to approximate periodic functions with sines and cosines.

7.1 Fourier Series

A function $f(x)$ that is periodic with period $2L$ can be written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

where the **Fourier coefficients** are given by

$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos \frac{n\pi x}{L}\, dx, \qquad b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin \frac{n\pi x}{L}\, dx$$

Remark. The term $a_0/2$ is the average value of f over one period.

Dirichlet Conditions

The Fourier series converges to $f(x)$ at all points where f is continuous, provided:

(i) f is periodic,

(ii) f is piecewise continuous on $[-L, L]$,

(iii) f has finitely many maxima and minima per period.

At a jump discontinuity at x_0, the series converges to the average *Remark.* Inverse trig functions cannot be represented by a Fourier series b/c they are not single-valued. Use Principle Branch workaround. $\frac{1}{2}\left[f(x_0^+) + f(x_0^-)\right]$.

Complex Form

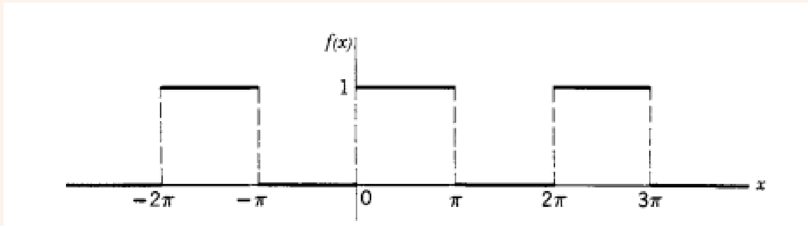
The Fourier series can equivalently be written as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n\, e^{in\pi x/L}, \qquad c_n = \frac{1}{2L} \int_{-L}^{L} f(x)\, e^{-in\pi x/L}\, dx$$

The real and complex coefficients are related by $c_n = \frac{1}{2}(a_n - ib_n)$ and $c_{-n} = \frac{1}{2}(a_n + ib_n)$.

Example 7.1: Square wave

Let $f(x)$ be the 2π-periodic square wave with $f(x) = 1$ for $0 < x < \pi$ and $f(x) = -1$ for $-\pi < x < 0$. Then $a_n = 0$ (odd function) and



$$b_n = \frac{2}{n\pi}(1 - \cos n\pi) = \begin{cases} \frac{4}{n\pi} & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

so

$$f(x) = \frac{4}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right)$$

7.2 Even and Odd Functions

Definition 7.1: Even and Odd Functions

A function is *even* if $f(-x) = f(x)$ and *odd* if $f(-x) = -f(x)$.

Key consequences for Fourier series on $[-L, L]$:

- **Even** f: all $b_n = 0$; the series contains only cosines.

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx$$

- **Odd** f: all $a_n = 0$; the series contains only sines.

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx$$

Useful identities: even $\times$ even $=$ even; odd $\times$ odd $=$ even; even $\times$ odd $=$ odd.

7.3 Half-Range Series

Given $f(x)$ defined only on $[0, L]$, we can extend it to $[-L, L]$ in two ways and obtain a *half-range* series:

- **Half-range cosine series** (even extension):

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{L}, \qquad a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx$$

- **Half-range sine series** (odd extension):

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{L}, \qquad b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx$$

Remark. The choice of extension affects convergence at the endpoints $x = 0$ and $x = L$. The two series are generally different functions outside $[0, L]$.

7.4 Parseval's Theorem

Theorem 7.1: Parseval's Theorem

If $f(x)$ has Fourier coefficients a_n, b_n on $[-L, L]$, then

$$\frac{1}{L}\int_{-L}^{L}[f(x)]^2\,dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)$$

Parseval's theorem equates the *energy* (mean-square value) of f to the sum of the squared amplitudes of its frequency components. It is useful for evaluating infinite series.

Example 7.2: A classic sum

The Fourier series of $f(x) = x$ on $[-\pi, \pi]$ gives $b_n = 2(-1)^{n+1}/n$. Applying Parseval's theorem:

$$\frac{1}{\pi}\int_{-\pi}^{\pi} x^2\,dx = \sum_{n=1}^{\infty}\frac{4}{n^2} \implies \frac{2\pi^2}{3} = 4\sum_{n=1}^{\infty}\frac{1}{n^2} \implies \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}$$

7.5 Fourier Transforms

For a non-periodic function $f(t)$ defined on $(-\infty, \infty)$, the Fourier series generalizes to the **Fourier transform pair**:

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t)\,e^{-i\omega t}\,dt \qquad \text{(forward transform)}$$

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\hat{f}(\omega)\,e^{i\omega t}\,d\omega \qquad \text{(inverse transform)}$$

Key Properties

Property	Statement
Linearity	$\widehat{af + bg} = a\hat{f} + b\hat{g}$
Time shift	$\widehat{f(t - t_0)} = e^{-i\omega t_0}\hat{f}(\omega)$
Frequency shift	$\widehat{e^{i\omega_0 t}f(t)} = \hat{f}(\omega - \omega_0)$
Differentiation	$\widehat{f'(t)} = i\omega\,\hat{f}(\omega)$
Scaling	$\widehat{f(at)} = \frac{1}{\lvert a\rvert}\hat{f}(\omega/a)$

Common Transform Pairs

$$e^{-a|t|} \longleftrightarrow \frac{2a}{a^2 + \omega^2}, \qquad a > 0$$

$$e^{-at^2} \longleftrightarrow \sqrt{\frac{\pi}{a}}\,e^{-\omega^2/(4a)}, \qquad a > 0$$

$$\delta(t) \longleftrightarrow 1, \qquad 1 \longleftrightarrow 2\pi\,\delta(\omega)$$

Remark. Different textbooks use different conventions for the factor of 2π and the sign in the exponent. Boas uses the convention above.

7.6 Convolution

Definition 7.2: Convolution

The *convolution* of two functions f and g is

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)\, g(t-\tau)\, d\tau$$

Theorem 7.2: Convolution Theorem

The Fourier transform of a convolution is the product of the transforms:

$$\widehat{f * g}(\omega) = \hat{f}(\omega) \cdot \hat{g}(\omega)$$

Equivalently, the transform of a product is (up to 2π) the convolution of the transforms:

$$\widehat{f \cdot g}(\omega) = \frac{1}{2\pi} \big(\hat{f} * \hat{g}\big)(\omega)$$

Properties of Convolution

- Commutative: $f * g = g * f$
- Associative: $f * (g * h) = (f * g) * h$
- Distributive: $f * (g + h) = f * g + f * h$

Remark. Convolution appears naturally in linear systems: if $h(t)$ is the impulse response of a system and $x(t)$ is the input, the output is $y(t) = (h * x)(t)$.

Remark. I will add given examples from the Final Review here after I am finished going through the chapters.

8 Ordinary Differential Equations

8.1 First-Order ODEs

8.2 Second-Order Linear ODEs

8.3 Variation of Parameters

8.4 Power Series Solutions

8.5 Frobenius Method

8.6 Bessel's Equation (intro)

9 Tensor Analysis

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9.4 Kronecker Delta and Levi-Civita Symbol

9.5 Isotropic Tensors

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10.2 Legendre Polynomials

10.3 Orthogonality and Expansion

10.4 Bessel's Equation

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10.6 Orthogonality of Bessel Functions

10.7 Applications

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11.2 Heat Equation

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12.4 Laurent Series

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12.6 Residue Theorem

12.7 Evaluation of Real Integrals

A Formula Sheet

Legendre Polynomials

Bessel Function Zeros

Gamma Function

Common Fourier Transforms

Vector Identities